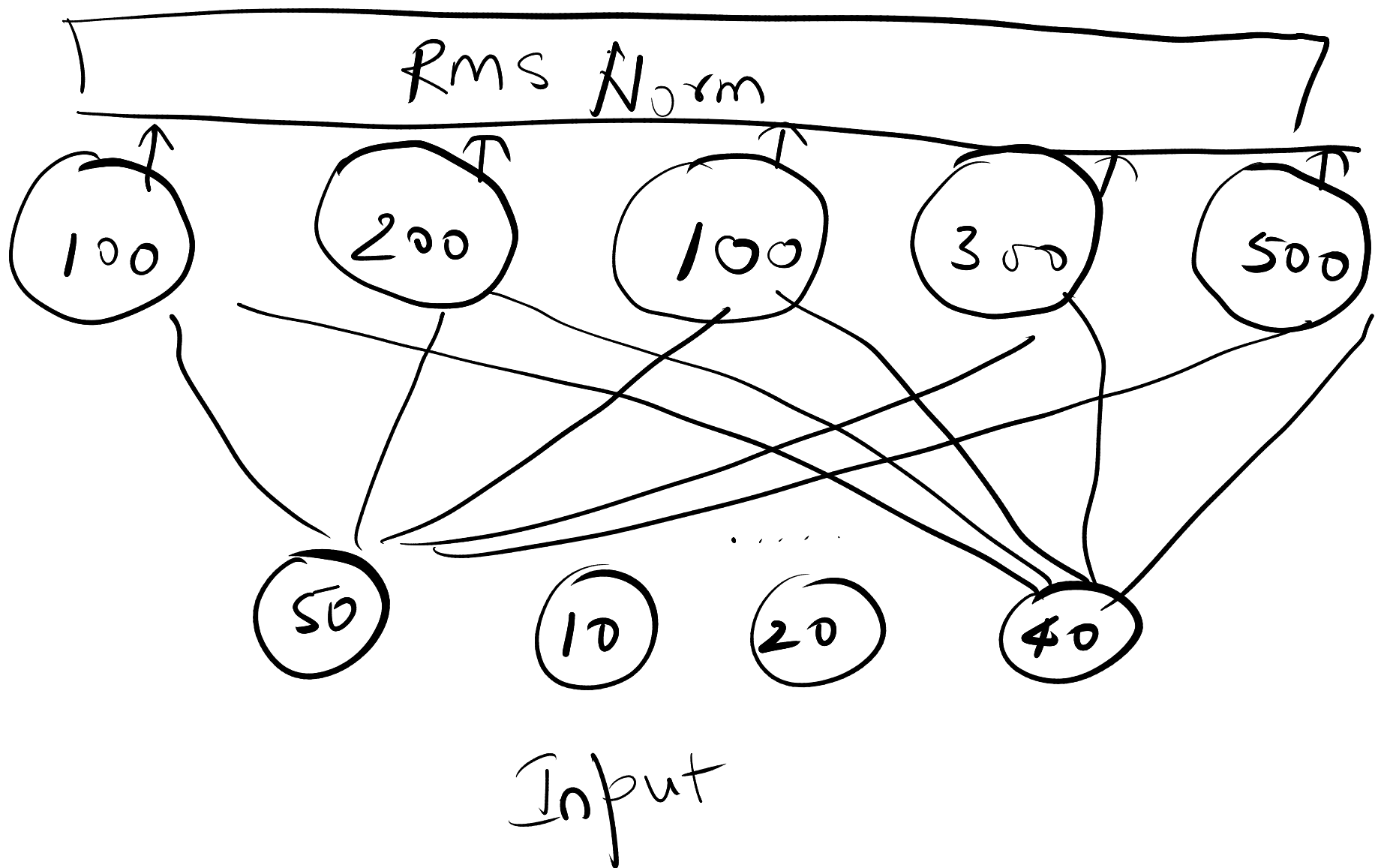


Session - 04 Advancements in LLM Architecture

$$\text{RMS}(a) = \sqrt{\frac{1}{n} \sum_{i=1}^n a_i^2}$$



Layer Norm :

$$\mu = \frac{1}{n} \sum_{i=1}^n a_i$$

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - \mu)^2}$$

$$\mu = \frac{1200}{5} = 240$$

$$\sigma = \sqrt{\frac{1}{5} [(-140)^2 + (-40)^2 + (-140)^2 + (60)^2 + (260)^2]}$$

$$\sigma = 149.66 \approx 150$$

$$\text{Layer Norm: } \gamma \left(\frac{x - \mu}{\sigma} \right) + \beta$$

$$= (1) \left(\frac{100 - \frac{240}{150}}{\frac{150}{150}} \right) + 0$$

$$= -0.933$$

⋮

RMS Norm :-

$$\begin{aligned} \text{RMS} &= \sqrt{\frac{1}{5} (100^2 + 200^2 + 100^2 + 300^2 + 500^2)} \\ &= 282.84 \approx 283 \end{aligned}$$

$$\text{RMS (a)} = \frac{100}{282.84} = 0.35$$

$$= \frac{200}{282.84} = 0.7071$$

$$= \frac{300}{282.84} = 1.06$$

$$= \frac{500}{282.84} = 1.77$$

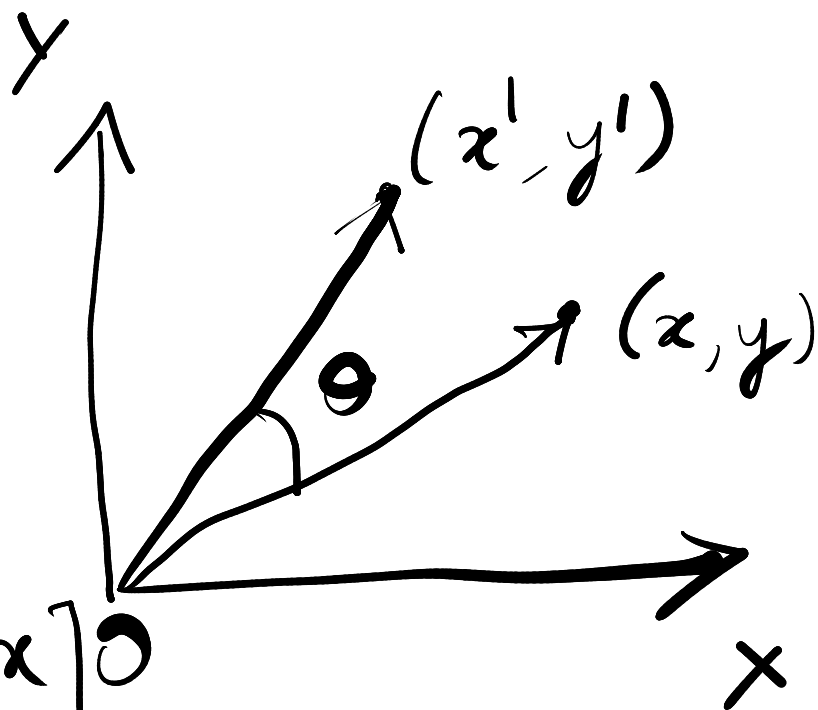
$$\text{Normalized RMS} = \underline{\underline{[0.35, 0.7071, 0.35, 1.06, 1.77]}}$$

QK-Norm:

$$\hat{q}_i = \frac{q_i}{\|q_i\|_2}$$

Basic Trigonometry

$$\begin{aligned} x' &= x \cos \theta - y \sin \theta \\ y' &= x \sin \theta + y \cos \theta \end{aligned}$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_{R(\theta)} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$R^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\stackrel{2}{=} \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix}$$

$$= R(-\theta)$$

$$\text{Score}(m, n) = q_m^T k_n$$

$$\tilde{q}_m = R(m\theta) q_m ; \quad \tilde{k}_n = R(n\theta) k_n$$

$$\text{Score}_{\text{RoPE}}(m, n) = \tilde{q}_m^T \tilde{k}_n$$

$$q_m^T k_n = \downarrow (R(m\theta) q_m)^T \cdot (R(n\theta) k_n)$$

$$\begin{aligned} R(\theta) = R(-\theta) &= q_m^T R(m\theta)^T \cdot R(n\theta) k_n \\ &= q_m^T R(-m\theta) R(n\theta) k_n \\ &= q_m^T R(\underbrace{(n-m)\theta}_{\text{relative position}}) k_n \end{aligned}$$

$R(\theta) \in$;

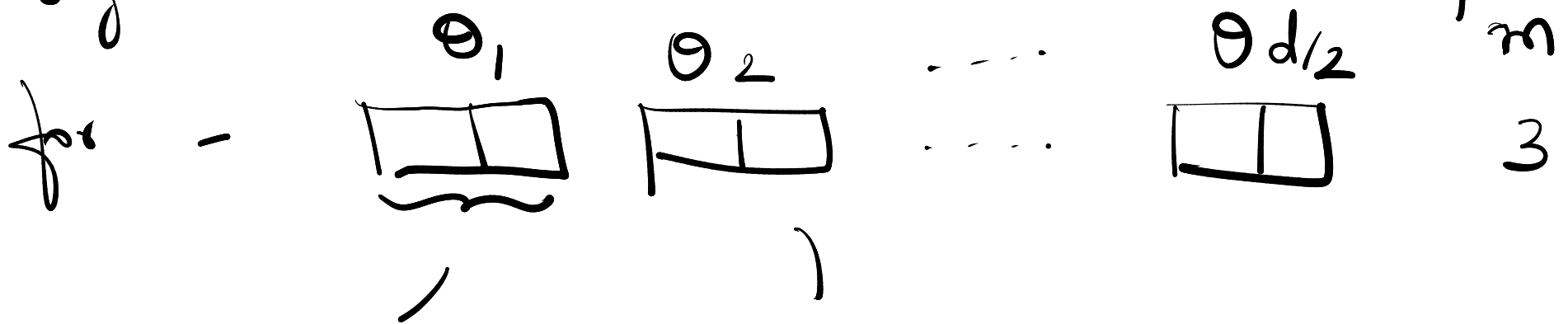
Each Vector $d/2$ Pairs

2D Pairs: $(x_1, x_2) (x_3, x_4) \dots$

Different rotation angle for each pair

$$\theta = 10000^{-2(i-1)/d}; i \in [1, 2, \dots, d/2]$$

Eg:-



$$\begin{bmatrix} \cos m\theta_1 & -\sin m\theta_1 \\ \sin m\theta_1 & \cos m\theta_1 \end{bmatrix} \begin{bmatrix} \cos m\theta_2 & -\sin m\theta_2 \\ \sin m\theta_2 & \cos m\theta_2 \end{bmatrix} \dots$$

$$R_{\theta, m}^d =$$

$$\begin{bmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \dots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \dots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \dots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{bmatrix}$$